

Why Did You Do That? An Economic Examination of the Effect of Extrinsic Compensation on Intrinsic Motivation and Performance

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Abstract: According to empirical evidence, extrinsic incentives often crowd out intrinsic motivation, thus reducing the effort choices of workers. This article presents a principal-agent model that incorporates insights from cognitive evaluation theory in order to show how the introduction of explicit incentives causes a discontinuous reduction in worker effort and why motivation crowding out occurs. Motivation crowding out is shown to be a rational response of the agent to the introduction of incentives when rewards are perceived as controlling, since controlled behavior does not provide intrinsic satisfaction. Extrinsic rewards could be perceived as controlling under two conditions. The first is when the object of an agent's intrinsic motivation is also the source of the agent's extrinsic compensation. The second is when the incentives offered to the agent are too large.

Keywords: Principal-agent problem, incentive compensation, intrinsic motivation, motivation crowding out

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I. Introduction

Although recognized by social psychologists (see deCharmes, 1968; Deci, 1975; Deci and Ryan, 1985), the idea that "providing extrinsic incentives for workers can be counterproductive, because it may destroy the worker's intrinsic motivation, leading to lessened levels of quality-weighted effort and lower net profits for the employer" is just a stylized fact within economics (Kreps, 1997, p. 360). Nevertheless, economists are accumulating increasing evidence for this effect, known as motivation crowding out (see Frey and Jegen, 2001). For example, Gneezy and Rustichini (2000) conducted experiments in which subjects were offered "employment" contracts that required them to answer simple questions or to perform simple tasks (e.g., collect donations). Some subjects were offered fixed wages for participating and then told to complete as many tasks as possible. Other subjects were offered a fee for participating but then an additional "incentive" payment based on their productivity. Gneezy and Rustichini found that higher incentive rates induced greater effort, but the effort of workers given only a fixed fee often exceeded the effort of workers paid an incentive wage. Fehr and Gächter (2000a) reported similar results from an experiment in which a principal offered a contract to agents to provide effort in exchange for payment. In some contracts the principal offered only fixed wages. In other contracts the principal offered a fixed wage but retained the right to punish (e.g., fine) shirking agents. Fehr and Gächter found that the (negative) incentive contracts, on average, elicited lower levels of effort from agents relative to the fixed wage contracts.

In spite of evidence that explicit incentives often crowd out intrinsic motivation and thus induce lower levels of effort, economists have been unable to provide convincing economic explanations for the crowding out effect (Kreps, 1997; Frey and Jegen, 2001). For example, Gneezy and Rustichini (2000) considered two possible explanations for their experimental findings. The first explanation was that agents are motivated by extrinsic and intrinsic factors and that the introduction of monetary compensation "crowds out" intrinsic motivation. Although they accepted that some form of crowding out seemed to occur, Gneezy and Rustichini rejected this explanation because it does not explain the discontinuous reduction in performance they observed when incentives are introduced. The second explanation was that the change from a fixed-fee to an incentive contract is associated with a change in agent perceptions of the contract because the contract is incomplete. An agent's "perception," in effect, fills in the missing parts (e.g., how much effort to supply). Without explicit incentives, agents perceive the contracts as embodying an implicit obligation on their part to provide effort, which obligation might be based on social norms of cooperation or reciprocity (see Fehr and Gächter, 2000b). However, when incentives are introduced the perception might be that effort should be rewarded according to the piece rate only, thus negating the need for a perception based on social norms. Although social norms may exist (see Ostrom, 2000) in support of intrinsic incentives that induce agents to supply effort when contracts are incomplete and explicit incentives are absent, this does not explain why the introduction of incentives changes the perceived obligation of workers to supply effort.

This paper presents a model illustrating *how* the combination of extrinsic and intrinsic motivations explains the discontinuous shift in worker effort when incentives are introduced. The model also helps explain *why* extrinsic incentives crowd out intrinsic motivation. In contrast to

extrinsic motivation, which is oriented toward obtaining a valued external reward, intrinsic drives "are the prototypes of autonomous or self-determined actions because they are performed out of interest and for the inherent satisfaction they yield. When intrinsically motivated, people *want* to engage in the activity; thus no external or intrapsychic prods, promises, or threats are required" (Ryan, Sheldon, Kasser, and Deci, 1996, p. 10; emphasis in original). In this context motivation crowding out means that the offering of extrinsic incentives to an agent represented by a utility function comprising both extrinsic and intrinsic components causes the agent to behave *as if* the intrinsic properties of his utility no longer factor in his decision-making processes – that is, as if his intrinsic motivation has been "crowded out," "neutralized," or "deactivated." This raises an important question: Is motivation crowding out the result of a conscious choice of the agent not to be intrinsically motivated, or does it reflect subconscious automatic processes beyond the relevant control of the agent?¹ Regardless of whether agents actually have or choose to have direct control over what motivates them, a model in which agents are assumed to maximize utility *as if* being intrinsically motivated is rationally determined will provide important insights into motivation crowding out. Indeed, unlike previous models that explain motivational crowding out by making *ad hoc* assumptions about the signs of partial derivatives (as in Frey, 1993 and 1994; Frey and Oberholzer-Gee, 1997), this paper shows how the crowding out of intrinsic motivation is a rational response of the agent to the introduction of incentives. This argument is founded on cognitive evaluation theory (CET) developed by Deci and Ryan (1980, 1985). According to CET, the basic factors

underlying intrinsic motivation are the psychological needs for autonomy and competence, so the effect of an event such as a reward depend on how it affects

¹ For instance, Sorrentino (1996, p. 619) depicts the two possibilities as follows: On the one hand, "motivation ... stems from conscious decision making, and nonconscious motives or cognitions have little relevance." On the other hand, "nonconscious memories and unintended thought play major roles in determining much of human behavior."

perceived self-determination and perceived competence. Events that allow need satisfaction tend to increase intrinsic motivation whereas those that thwart need satisfaction tend to decrease intrinsic motivation. CET proposes that rewards can be interpreted by recipients primarily as controllers of their behavior or, alternatively, as indicators of their competence. (Deci, Koestner, and Ryan, 1999a, p. 628)

In other words, if an agent perceives that an event supports his innate need for self-determination and competence, then even if the event provides extrinsic value (i.e., increases his extrinsic utility), his behavior vis-à-vis the event will also provide intrinsic satisfaction (i.e., increase his intrinsic utility). Hence, it would be rational for him to behave *as if* extrinsically and intrinsically motivated. On the other hand, if an agent perceives that an event is controlling rather than the result of self-determined behavior, then it will not provide intrinsic satisfaction. Accordingly, it would be rational for him to behave *as if* extrinsically motivated only (i.e., intrinsic utility is not factored in). This characterizes motivation crowding out.

Therefore, the key to understanding motivation crowding out is through a model in which an agent's utility reflects either (a) extrinsic and intrinsic factors if extrinsic rewards are not perceived as controlling, or (b) extrinsic factors only if extrinsic rewards are perceived as controlling. Extrinsic rewards might be perceived as controlling under two conditions. The first condition is when the size of the reward is large. Empirical evidence suggests that reward size is negatively related to intrinsic motivation (e.g., Newman and Layton, 1984). The reason is that the processes by which an agent rationalizes his behavior may become so overwhelmed by the salience (e.g., size) of extrinsic rewards that he is rationally compelled to attribute his behavior to the compensation rather than to his intrinsic preferences. This is known as the "overjustification effect" (see Lepper, Greene, and Nisbett, 1973; Lepper, 1981). This idea was recognized by Kreps (1997, p. 362) who said:

Turning revealed preference on its head, the idea is that when a person performs some act, he looks for rationales that justify his actions. Specifically, if an

employee undertakes some effort without the spur of some extrinsic incentive, he will rationalize his efforts as reflecting his enjoyment of the task. And since he enjoys it, he works harder at it. But if extrinsic incentives are put in place, he will attribute his efforts to those incentives, developing a distaste for the required effort.

Similarly, deCharmes (1968, p. 328) stated:

As a first approximation, we propose that whenever a person experiences himself to be the locus of causality for his own behavior ..., he will consider himself to be intrinsically motivated. Conversely, when a person perceives the locus of causality for his behavior to be external to himself ..., he will consider himself to be extrinsically motivated.

In other words, large rewards may compel an agent to acknowledge that the "locus of causality" is external rather than internal and is intended as a means of control. This would undermine the agent's inherent need for autonomy and hence provide no intrinsic satisfaction for engaging in the "controlled" behavior (see Deci, Koestner, and Ryan, 1999b). The result is that the agent would behave based only on the extrinsic utility created, resulting in motivation crowding out.

The second condition is that the object of an agent's intrinsic motivation is also the source of the rewards. For example, if an agent is initially intrinsically motivated to act in the interest of a principal who also pays the worker's wages, then the introduction of incentives by the principal could be perceived by the agent as an attempt at manipulation. This perception would undermine the agent's need for autonomy, thus causing motivation crowding out. However, if the agent is motivated by generalized norms (such as "provide an honest day's work for an honest day's pay") which by nature are external to the more narrow interests of the principal, then extrinsic incentives introduced by the principal might not be perceived as controlling but rather as an affirmation of competence, thus supporting rather than crowding out intrinsic motivation. This idea is consistent with empirical evidence. Although studies suggest that expected tangible rewards generally reduce intrinsic motivation (see Deci, Koestner, and Ryan, 1999a), rewards

offered for exceeding a norm or a minimum level of effort not only do not cause motivation crowding out but also often have positive effects (see Cameron, Banko, and Pierce, 2001).

Therefore, the objective of this paper is to show how the size of compensation and the object on which intrinsic motivation is placed interact so as to induce an agent maximizing utility comprising extrinsic and intrinsic elements to behave as if the intrinsic motivation is crowded out. In doing so, this paper supplements Murdock's (2002) article showing how incentive contracts in a principal-agent framework optimally respond to the presence of agent intrinsic motivation. In his paper, intrinsic motivation complements an implicit contract between a firm and worker in that the firm agrees to implement projects generating positive intrinsic utility for the agent but negative expected returns for the firm. The firm gains in the long run, however, because the intrinsic motivation of the worker increases the effect of effort incentives attached to other projects that are profitable, thus increasing the overall returns of the firm. In contrast, this paper shows that under certain circumstances the use of extrinsic incentives could backfire by reducing the effort choices of agents interested in generating profits for the firm.

The paper begins with a simple principal-agent model in which an agent's utility incorporates both extrinsic and intrinsic elements. Initially it is assumed that the object of the agent's intrinsic motivation is also the source of his compensation in that the agent is intrinsically motivated to generate profits for the principal. The model is used to demonstrate how the introduction of incentives could result in a discontinuous reduction in worker effort if the explicit incentives crowd out intrinsic motivation. The model is then used to show why the size of the incentive compensation is important in understanding why an agent intrinsically motivated to act in the interest of a principal would behave as if motivated by extrinsic factors only. A stylized numerical example is provided to illustrate the points raised in the analysis. The model is then

revised to reflect an agent's intrinsic motivation that is tied to a generalized norm of behavior rather than to the interests of a principal. This revision illustrates why the object on which intrinsic motivation is tied is important in understanding why motivation crowding out occurs. The paper concludes with a discussion of the findings.

II. Model

Suppose a principal hires an agent to supply effort, e , in exchange for compensation, w . If p is the revenue generated per unit of effort, then the principal's profits are $pe - w$. If the worker's disutility of effort is e^2 , then his extrinsic utility is $w - e^2$. Optimally, the worker should provide effort $e^* = p/2$, which maximizes social welfare (profits plus utility).

A. Explicit Incentives and the Discontinuous Reduction of Effort

How do we account for evidence that the introduction of incentives results in a discontinuous reduction of worker effort? In order to answer this question, a model of utility is required which contains both extrinsic and intrinsic sources of utility. Thus, suppose the agent chooses effort, e , to maximize the following utility function comprising both extrinsic and intrinsic elements:

$$U = \bar{w} + re - e^2 + I\delta s. \quad (1)$$

The extrinsic motivation of the agent is $\bar{w} + re - e^2$, which consists of compensation from a fixed fee, $\bar{w}$, and an incentive payment, r , net of effort cost e^2 . The agent's intrinsic motivation is represented as $I\delta s$. The variable I is an indicator of the presence or absence of the agent's intrinsic motivation, such that $I = 1$ if the agent is intrinsically motivated and $I = 0$ if his

intrinsic motivation is crowded out. Following Murdock (2002), the parameter δ represents the intensity of the agent's intrinsic motivation for some object, s , where an agent with a high δ will have a stronger motivation toward s than an agent with a low δ .

For simplicity, assume $p > r$ (so that incentives are marginally profitable to the principal) and $\delta > 0$ (so that the agent is motivated to be productive rather than destructive). Moreover, assume a two-period framework. In period one a principal offers an employment contract to an agent randomly drawn from a population consisting of workers whose reservation wage is zero and who are identical in their abilities, but are different in the size of δ defining the strength of the intrinsic motivation they feel, so that agents with higher δ s are more intrinsically motivated than agents with lower δ s, other things being equal. Assume further that the level of δ is private information and not knowable to the principal at any cost. Therefore, we are abstracting away from the principal's problem of identifying agents with high δ s. Finally, assume that the intensity of an intrinsically motivated agent's innate motivation, δ , is fixed; that is, we are abstracting away from the problem of how to increase or decrease at the margin the intrinsic motivation of workers.² The contract offered to the agent consists of a fixed wage, $\bar{w}$, and an incentive rate, $r \geq 0$, in exchange for effort, e , chosen by the agent. In period two the agent makes the effort choice and the principal compensates the worker, which occur simultaneously.

Assume initially the object of the agent's intrinsic motivation is the principal (e.g., the agent is motivated to generate profits for the principal) so that $s = pe - \bar{w} - re$ (the principal is

² To be clear here, the question of interest is only whether or not the agent would behave as if intrinsically motivated (i.e., whether $I = 1$ rather than $I = 0$), not how intrinsically motivated he would be (i.e., how large or small δ is). This is consistent with the assessment of the empirical evidence by Ryan *et al* (1996, p. 9) who conclude that the "amount or level of motivation does not necessarily differ when people are autonomous versus controlled, but the type or orientation of motivation does."

also assumed to be interested in securing maximized profits). Accordingly, equation (1) is rewritten as

$$U = \bar{w} + re - e^2 + I\delta(pe - \bar{w} - re). \quad (1')$$

Given equation (1') and the assumptions of the model, the agent's optimal effort choice, $\hat{e}$, obtained by maximizing (1') with respect to effort, is

$$\hat{e} = \frac{I\delta(p - r) + r}{2}. \quad (2a)$$

Now consider two different stylized scenarios. In the first the agent is offered only a fixed fee "for showing up," but he is also intrinsically motivated to supply effort. In this case $r = 0$ and $I = 1$ so that the agent supplies effort $\hat{e} = \delta p / 2 \equiv \hat{e}_1$. In the second the agent is offered a fixed fee and an incentive, r , for effort, but he has no intrinsic motivation to provide effort above what the incentive offers. In this case $r > 0$ and $I = 0$ so that the agent supplies effort $\hat{e} = r / 2 \equiv \hat{e}_2$. Note that in this second case effort increases as the piece rate increases (the optimal effort-incentive profile is positive). These two scenarios conform to the stylization of the motivation crowding out problem in the literature in that workers are often intrinsically motivated when no incentive compensation is offered (i.e., $r = 0$ and $I = 1$) but are not intrinsically motivated when incentives are offered (i.e., $r > 0$ and $I = 0$). Equation (2b) summarizes the effort choice of the agent under each of these two scenarios.

$$\hat{e} = \begin{cases} \delta p / 2 \equiv \hat{e}_1 & \text{if } r = 0, I = 1 \\ r / 2 \equiv \hat{e}_2 & \text{if } r > 0, I = 0 \end{cases}. \quad (2b)$$

According to the experimental evidence, effort supplied under a fixed-fee contract with an intrinsically motivated agent will often equal or exceed effort supplied by an agent not intrinsically motivated when governed by an incentive contract (i.e., $\hat{e}_1 \geq \hat{e}_2$). This would be true

when $\delta p / 2 \geq r / 2$, or when $\delta \geq r / p$, suggesting that the intrinsic motivation of the agent must be sufficiently "strong." In other words, when $\delta \geq r / p$, then effort provided by the agent when no extrinsic incentives are offered (equal to $\hat{e}_1$) will not be less than effort provided by the agent when incentives are offered but the agent's intrinsic motivation is absent (equal to $\hat{e}_2$). If incentives are subsequently introduced and cause the agent's intrinsic motivation to be crowded out so that $I = 0$, however, then the effort choice of the agent discontinuously declines from $\hat{e}_1$ to $\hat{e}_2$. Thereafter, any increase in piece rates results in increased effort (i.e., as r increases, $\hat{e}_2$ increases continuously). This is illustrated in Figure 1.

[Figure 1 about here]

B. Motivation Crowding Out When Duty is Directed Toward a Principal

Why might the introduction of an explicit incentive be associated with an apparent disappearance of an agent's intrinsic motivation? That is, why might positive incentives ($r > 0$) be associated with agents behaving as if $I = 0$? This question is especially germane when one recognizes that the agent's utility is increasing in δ . Wouldn't the agent be better off being intrinsically motivated in addition to being paid incentive wages (i.e., "prefer" that $I = 1$ rather than $I = 0$ when $r > 0$) since intrinsic motivation adds to the agent's utility? In order to understand why an agent with utility comprising extrinsic and intrinsic elements would choose effort as if intrinsic motivation were crowded out, we must examine the utility of the agent when $I = 0$ (intrinsic motivation crowded out) and $I = 1$ (positive intrinsic motivation) under the two cases of $r = 0$ (no incentive compensation) and $r > 0$ (positive incentive compensation).

Simply, will being intrinsically motivated or not being intrinsically motivated give the agent greater utility in the absence and presence of incentive compensation?

First, consider the case in which no incentive pay is offered so that $r = 0$. If the agent's intrinsic motivation is crowded out so that $I = 0$, then the utility of the agent is

$$U(\hat{e} | r = 0, I = 0) = \bar{w}, \quad (3a)$$

which corresponds to an effort choice of $\hat{e} = 0$. If, however, the agent is intrinsically motivated so that $I = 1$, then the utility of the agent is

$$U(\hat{e} | r = 0, I = 1) = \frac{\delta^2 p^2 - 4\delta\bar{w} + 4\bar{w}}{4}, \quad (3b)$$

which corresponds to an effort choice by the agent of $\hat{e}_1 (= \delta p / 2)$. The agent will have greater utility when intrinsically motivated when equation (3b) is not less than (3a), or when

$$\delta \geq \frac{4\bar{w}}{p^2} \equiv \delta_{FW}. \quad (4)$$

Equation (4) defines the threshold determining whether an agent will be intrinsically motivated to provide effort for a principal when offered only a fixed wage (FW) contract. When $\delta \geq \delta_{FW}$, or when an agent's intrinsic motivation to act in the interest of the principal is relatively high, then he will have higher utility when intrinsically motivated. However, when $\delta < \delta_{FW}$, or when the agent's intrinsic motivation is relatively low, then he will have higher utility by choosing effort as if not intrinsically motivated such that $I = 0$; that is, intrinsic motivation would be crowded out. Observe that the threshold δ_{FW} increases in the amount of the fixed wage but decreases in the revenue generated per unit of effort. That is, the larger the fixed wage offered to the agent, the higher the agent's innate intrinsic motivation would have to be in order to induce him to provide effort, other things being equal. Simply, if the fixed wage offered to the

agent is too large, the agent may decide it is in his interest to take the money and run (i.e., provide no effort). According to experimental (and anecdotal) evidence, however, this does not seem to be the case. Most individuals, when offered a fixed wage but no incentive compensation, seem to be willing to work as if intrinsically motivated (i.e., $I = 1$), suggesting that $\delta \geq \delta_{FW}$ with effort corresponding to $\hat{e}_1$.

Now, consider the case in which incentive pay is offered so that $r > 0$. If the agent is not intrinsically motivated so that $I = 0$, then the utility of the agent is

$$U(\hat{e} | r > 0, I = 0) = \frac{r^2 + 4\bar{w}}{4}, \quad (5a)$$

which corresponds to an effort choice of $\hat{e}_2 (= r/2)$. If, however, the agent is intrinsically motivated so that $I = 1$, then the utility of the agent is

$$U(\hat{e} | r > 0, I = 1) = \frac{\delta^2 (p - r)^2 + 2\delta(pr - r^2 - 2\bar{w}) + r^2 + 4\bar{w}}{4}, \quad (5b)$$

which corresponds to effort choice $\hat{e} = \frac{\delta(p - r) + r}{2}$. The agent will have greater utility when intrinsically motivated when equation (5b) is not less than (5a), or when

$$\delta \geq \frac{4\bar{w} - 2r(p - r)}{(p - r)^2} \equiv \delta_{IC}. \quad (6)$$

Equation (6) defines the threshold determining whether an agent will be intrinsically motivated when offered an incentive compensation (IC) contract. If $\delta \geq \delta_{IC}$, or if the agent's innate intrinsic motivation toward the principal is sufficiently strong, then even when incentive compensation is offered the agent will have higher utility being intrinsically motivated when choosing the level of effort to provide for the principal. If, however, the agent's innate intrinsic motivation is too low in the sense that $\delta < \delta_{IC}$, then the agent will derive greater satisfaction if

his intrinsic motivation is crowded out in that he chooses effort based solely on the size of the extrinsic incentive, r . In effect, the incentive compensation results in a rational "deactivation" of the agent's intrinsic motivation. Like the case in which no incentive compensation is offered, the threshold δ_{IC} increases in the amount of the fixed wage. How the threshold changes as p and r increase, however, is indeterminate.

[Figure 2 about here]

Figure 2 summarizes the utility and corresponding effort choice of the agent as described in the preceding discussion. The experimental evidence suggests that agents generally behave as if intrinsically motivated when a fixed wage but no incentive rate is offered, suggesting that $\delta \geq \delta_{FW}$. This corresponds to effort choice $\hat{e}_1$ in cell III of Table 2. Therefore, assume for the following discussion that the $\delta \geq \delta_{FW}$ condition is satisfied so that the agent is providing effort $\hat{e}_1$. Suppose now the agent is offered an incentive. Will $\delta \geq \delta_{IC}$ so that effort in cell IV is selected, or will $\delta < \delta_{IC}$ so that effort $\hat{e}_2$ in cell II is selected? The experimental evidence suggests that when workers are offered incentive compensation, many who would have been or who were already intrinsically motivated when no incentives were paid subsequently act as if they are not intrinsically motivated, suggesting that $\delta < \delta_{IC}$ rather than $\delta \geq \delta_{IC}$. Why would the introduction of incentive compensation imply that effort in cell II is observed rather than effort in cell IV? Why would the introduction of an incentive imply that $\delta < \delta_{IC}$ rather than $\delta \geq \delta_{IC}$? To answer these questions we must determine whether and, if so, when, it is possible for

$\delta_{FW} \leq \delta < \delta_{IC}$. This expression would be true when $\frac{4\bar{w}}{p^2} < \frac{4\bar{w} - 2r(p-r)}{(p-r)^2}$, or when

$$r > \frac{p(p^2 - 4\bar{w})}{p^2 - 2\bar{w}} \equiv \bar{r} . \quad (7)$$

Equation (7) says that if the incentive rate offered by the principal is "too high," then some agents intrinsically motivated to provide effort in the absence of incentive compensation (i.e., with $\delta \geq \delta_{FW}$) will no longer be intrinsically motivated when incentive compensation is offered (i.e., $\delta < \delta_{IC}$). That is, their intrinsic motivation would be crowded out by the extrinsic incentives. If, however, $r \leq \bar{r}$, then $\delta_{FW} \geq \delta_{IC}$. Therefore, agents who would be intrinsically motivated when no incentive compensation is offered (i.e., $\delta \geq \delta_{FW}$) will also be intrinsically motivated when incentives are offered, since $\delta \geq \delta_{FW}$ implies $\delta > \delta_{IC}$ in this case. In other words, for some agents the question of why the introduction of extrinsic compensation "crowds out" intrinsic motivation and thus results in lower effort levels depends in part on how large the incentive rate is. For example, although nominal or "symbolic" (e.g., verbal) rewards would not likely have an adverse effect on intrinsic motivation, large tangible incentives designed to elicit higher levels of effort may be sufficient to erode intrinsic motivation, a prediction consistent with empirical evidence (see Deci, Koestner, and Ryan, 1999a; Frey and Jegen, 2001; Cameron, Banko, and Pierce, 2001). As explained above, a reason that "large" rewards crowd out intrinsic motivation is that they might be perceived by agents as controlling and, if so, would provide no intrinsic satisfaction to the agent. Therefore, his behavior would likely reflect extrinsic satisfaction only.

Explicit incentives need not be large in an absolute sense before motivation crowding out occurs. Even small monetary incentives could induce agents to behave as if not intrinsically motivated depending on how valuable the agent's efforts are for a principal, the size of the fixed wage, the agent's innate intensity of intrinsic motivation, and other factors. For instance, observe

that the incentive rate threshold, $\bar{r}$, increases in the revenue generated per unit of effort, p ; that is,

$$\frac{\partial \bar{r}}{\partial p} = \frac{(p^2 - \bar{w})^2 + 7\bar{w}^2}{(p^2 - 2\bar{w})^2} > 0. \quad (8a)$$

In other words, the more "valuable" or important the agent's effort is to the principal, as measured by the revenue generated by the worker's actions, the higher is the incentive rate the principal can offer the agent before motivation crowding out occurs, other things being equal. This is intuitive. Workers who are intrinsically motivated to take actions that benefit a principal will "naturally" be pleased when their efforts actually benefit the principal. Accordingly, it would take a larger incentive before these workers will experience the motivation crowding out effect compared to workers who provide effort resulting in only a small monetary gain to the principal.

Observe further that $\bar{r}$ decreases in the amount of the fixed wage, $\bar{w}$; that is,

$$\frac{\partial \bar{r}}{\partial \bar{w}} = -\frac{2p^3}{(p^2 - 2\bar{w})^2} < 0. \quad (8b)$$

This says that as the fixed wage increases, the minimum incentive rate at which motivation crowding out occurs will decline, suggesting that it is really total compensation, not just the incentive rate, which is important for motivation crowding out. Agents paid a relatively low fixed wage will not necessarily be rationally compelled to ignore their intrinsic interests when low or even moderate levels of incentives are offered. However, when agents are paid very large fixed wages, then even small incentives would produce the motivation crowding out effect.

C. Stylized Example of Motivation Crowding Out

In order to understand more clearly the relationship between compensation and intrinsic motivation, Table 1 presents a highly stylized example using hypothetical parameter values of

the model. The example is designed to illustrate how increasing levels of compensation might affect an agent's intrinsic motivation, effort choices and utility. For this example we are interested in those situations in which an agent already intrinsically motivated when no incentives are offered (i.e., when $\delta \geq \delta_{FW}$) will not be intrinsically motivated when incentives are offered (i.e., when $\delta < \delta_{FW}$); we are not interested in those cases in which agents who are not intrinsically motivated in the absence of incentives become intrinsically motivated when incentives are offered, defined as motivation crowding in (which is a theoretical possibility of this model).

[Table 1 about here]

Using the table one can compare the utility of the agent with and without intrinsic motivation for different levels of wages and incentive rates. In this example, the fixed wage, $\bar{w}$, is equal either to a low of 300 (e.g., "dollars") or to a high of 600 (see column 1). The revenue per unit of effort, p , is equal to a constant 70. The incentive rate, r , given in column 3, ranges from zero to low ($r = 20$) to moderate ($r = 50$) and finally to high ($r = 65$). The agent's innate intrinsic motivation, δ , is either low (equal to 0.3) or high (equal to 0.7), unless the agent is not intrinsically motivated ($I = 0$), in which case intrinsic motivation is zero (see column 4). The table also provides calculations for the minimum δ thresholds when fixed wages only are offered and when incentive compensation is included (δ_{FW} and δ_{IC} , respectively), the minimum incentive rate before intrinsic motivation is crowded out, $\bar{r}$, and the utility and effort choice of the agent (see columns 5 through 9, respectively).

For example, suppose $\bar{w} = 300$ but no incentive rate is offered. Row 1 gives the agent's utility and effort choice (see columns 8 and 9) when the agent is not intrinsically motivated (i.e., when $I = 0$). Row 2 provides these figures in the situation in which the agent is intrinsically motivated (i.e., when $I = 1$ and $\delta = .3$). Observe that the agent has higher utility by being intrinsically motivated; the agent's utility when not intrinsically motivated is 300, while his utility when intrinsically motivated is 320.25 (compare column 8, rows 1 and 2). Even though the agent's innate intrinsic motivation is relatively low, it still exceeds the minimum needed to induce effort (i.e., $\delta = .3 \geq .24 = \delta_{FW}$). Suppose now that a low incentive payment is offered to the agent (e.g., $r = 20$). Row 3 calculates the agent's utility and corresponding effort choice in the case in which he is not intrinsically motivated, while row 4 gives the agent's effort choice and utility in the case in which he is intrinsically motivated. Observe that the agent's utility when intrinsically motivated exceeds his utility when not intrinsically motivated (compare column 8, rows 3 and 4). Therefore, again, the agent has higher utility when intrinsically motivated. Indeed, this is also true when the incentive rate is moderate (e.g., $r = 50$). However, if the incentive rate is high (e.g., $r = 65$), then the agent has higher utility by choosing effort as if not intrinsically motivated rather than intrinsically motivated (compare column 8, rows 7 and 8). The incentive compensation crowds out the agent's intrinsic motivation because the incentive rate of 65 is "too high." Nonetheless, even though intrinsic motivation is crowded out in this case, the high incentive rate is sufficient to induce the agent to supply effort in excess of the effort he supplied when intrinsically motivated but paid a low or moderate incentive. Effort chosen when the agent is paid an incentive rate of 65 is equal to 32.50, which exceeds the effort choice of 28.00 obtained with an incentive rate of 50 and the agent acting from intrinsic motivation. In other

words, principals can compensate for a lack of intrinsic motivation by paying their workers enough (see Gneezy and Rustichini, 2000; Gneezy, 2003).

In the case in which the fixed wage is relatively high (e.g., $\bar{w} = 600$), the agent will be intrinsically motivated to provide effort only when his intrinsic motivation is also relatively high (i.e., $\delta \geq .49$). Otherwise the agent will provide no effort. Moreover, the introduction of a low incentive rate will not necessarily crowd out the intrinsic motivation for an agent in this case since $\delta = .7 > \delta_{IC} = .16$ (compare column 8, rows 11 and 12), but moderate to high incentive rates will crowd out intrinsic motivation. That is, the agent will have higher utility by choosing effort when not intrinsically motivated when the fixed wage is high and the incentive rate is moderate to high (compare column 8, row 13 to row 14, or row 15 to row 16).

D. Intrinsic Motivation When Duty is Directed Toward a Social Norm

In the preceding model it was assumed that the object of an agent's intrinsic motivation was also the source of the agent's extrinsic rewards in that the agent was assumed to derive satisfaction from acting in the interest of the principal and the principal was assumed to have an interest in maximizing profits. Suppose now that rather than being directed toward a principal, the object of an agent's intrinsic motivation is a generalized social norm, such as "provide an honest day's work for an honest day's pay" or "honor your contractual obligations." This could be interpreted to mean that the agent believes he should provide at least some minimal level of effort, $\bar{e}$, when employed by the principal, so that $s = e - \bar{e}$. Thus, instead of an objective function defined by equation (1'), the agent's objective function would be represented as

$$U = \bar{w} + re - e^2 + I\delta(e - \bar{e}). \quad (9)$$

In this formulation the utility of the agent is increased when $e > \bar{e}$ and is decreased when $e < \bar{e}$, thus giving the agent an incentive to adhere to the social norm, other things being equal. How does the agent know what the minimum level of effort should be? This minimum effort could be specified in an employment contract, in which case the social norm to which the agent would have an interest in following might be interpreted as "honor your contractual obligations." But this need not be the case. The minimum effort could be defined by a social norm such as "provide an honest day's work for an honest day's pay."

Given equation (9), the agent's optimal effort choice is

$$\tilde{e} = \frac{I\delta + r}{2}. \quad (10)$$

Consider now the utility of the agent under the cases of $I = 0$ or $I = 1$ when $r = 0$, and $I = 0$ or $I = 1$ when $r > 0$. First, suppose no incentive compensation is offered so that $r = 0$. If the agent is not intrinsically motivated to follow the social norm, then his utility is

$$U(\tilde{e} | r = 0, I = 0) = \bar{w}, \quad (11a)$$

which corresponds to an effort choice of $\tilde{e} = 0$. If, however, the agent is intrinsically motivated, then the agent's utility is

$$U(\tilde{e} | r = 0, I = 1) = \frac{\delta^2 - 4\delta\bar{e} + 4\bar{w}}{4}, \quad (11b)$$

which corresponds to effort choice $\tilde{e} = \delta/2$. The agent will have higher utility by behaving as if intrinsically motivated when equation (11b) is not less than equation (11a), or when

$$\delta \geq 4\bar{e} \equiv \tilde{\delta}_{FW}. \quad (12)$$

Suppose now the principal offers an incentive wage to the agent so that $r > 0$. In this case if the agent is not intrinsically motivated, then his utility is

$$U(\tilde{e} \mid r > 0, I = 0) = \frac{r^2 + 4\bar{w}}{4}, \quad (13a)$$

which corresponds to effort choice $\tilde{e} = r/2$. If, on the other hand, the agent is intrinsically motivated when incentive compensation is provided, then his utility is

$$U(\tilde{e} \mid r > 0, I = 1) = \frac{\delta^2 - 4\delta\bar{e} + 2\delta r + r^2 + 4\bar{w}}{4}, \quad (13b)$$

which corresponds to effort $\tilde{e} = \frac{\delta + r}{2}$. Equation (13b) would not be less than equation (13a), or the agent will have greater utility when intrinsically motivated to adhere to the social norm when

$$\delta \geq 4\bar{e} - 2r \equiv \tilde{\delta}_{IC}. \quad (14)$$

Because the incentive rate is assumed to be non-negative, the threshold level of innate intrinsic motivation necessary to induce intrinsic motivation when incentive are offered will always be lower than the threshold required when only fixed compensation is offered; that is, $\tilde{\delta}_{FW} \geq \tilde{\delta}_{IC}$. Therefore, if the agent is already intrinsically motivated to work when there are no extrinsic incentives to providing higher levels of effort (i.e., $\delta \geq \tilde{\delta}_{FW}$), then intrinsic motivation would not be crowded out when incentive compensation is offered, since $\delta \geq \tilde{\delta}_{FW}$ implies $\delta \geq \tilde{\delta}_{IC}$ in this case.

III. Concluding Discussion

This paper presented a model illustrating how and why the introduction of incentives might result in the crowding out of the intrinsic motivation of workers. The model utilizes insights from cognitive evaluation theory which posits that events perceived as controlling would induce agents not to be intrinsically motivated since controlled behavior does not provide

intrinsic satisfaction. The model shows that two factors are important in understanding how an agent might perceive the introduction of an explicit incentive as controlling – the object on which intrinsic motivation is tied and the size of the incentive and fixed compensation. Specifically, if the object on which the agent's motivation is tied also supplies the extrinsic reward, and if that reward is large enough, then the agent will perceive the incentive as controlling, the result being motivation crowding out.

This suggests the following testable hypotheses: First, agents motivated to advance a principal's interests would be more likely to experience the motivation crowding out effect than workers intrinsically motivated to adhere to generalized social norms. For instance, the more loyal a worker is to a particular principal, other things being equal, the more likely incentives will result in motivation crowding out because incentives offered to loyal workers might be perceived as controlling. This prediction has some empirical validity (see Barkema, 1995).

Second, the introduction of relatively low incentives may not necessarily result in the motivation crowding out effect if the fixed pay offered to agents is also relatively low. For instance, "symbolic" incentives would not necessarily result in motivation crowding out (see Frey and Jegen, 2001) and could in fact increase effort because it signals appreciation or recognition for effort expended. However, when fixed compensation is large, then even small incentives could result in motivation crowding out. The key here has to do with the size of the total compensation paid to workers vis-à-vis the benefits worker efforts provide the principal and the agent's innate level of intrinsic satisfaction. If total compensation is too large, then the salience of the extrinsic reward might be so overwhelming that the agent is rationally compelled to perceive the compensation as a mechanism of control, thus resulting in motivation crowding out. This suggests that an agent's effort supply curve might be S-shaped in the sense that small

incentives increase effort because they are not perceived as controlling, larger compensation reduces effort because of its negative effect on intrinsic motivation, but further increases in compensation increase effort in the standard economic sense.³

Of course, an agent with a high innate intrinsic intensity (δ in this paper) would not necessarily experience the motivation crowding out effect. This is consistent with some empirical evidence that finds that activities with high inherent interest motivation are not always negatively affected by extrinsic rewards (see Cameron, Banko, and Pierce, 2001). Consequently, the practical relevance of motivation crowding out depends fundamentally on how strong an agent's innate internal motivation is and what that motivation is tied to. This implies that principals have an interest in expending resources searching for workers who have high intrinsic motivations. However, this paper has also shown that understanding the object on which an agent's intrinsic motivation is placed is just as important in principal-agent relationships as the development of optimal incentive contracts. For example, indoctrinating workers on the importance of acting in the firm owners' interests when incentives are also used to help align the interests of owner and worker may be counterproductive. The reason is that workers may "see through" attempts by firm owners to maximize profits as enriching the owners at the expense of the firm and thus perceive incentives as a blatant form of control. According to Miller (2001), "Holmstrom's impossibility result has supplied an alternative vision of opportunism – opportunism on the part of owners themselves. It is an opportunism that follows from their ownership of residual profits, and it paradoxically puts them at odds with overall efficiency of the firm" (p. 329). The idea is that under certain circumstances firm owners could increase their residual returns by fostering inefficiencies within the organization. If workers expect this

³ This is illustrated in rows 10, 12, 13, and 15 of Table 1. See Le Grand (2003) for a complementary discussion.

because of employer attempts to control worker effort by means of extrinsic rewards, workers may be less inclined to "trust" firm owners who claim that certain activities or programs are for the good of the firm, thus resulting in lower returns for the firm owners. This might explain why Fehr and Gächter (2000a) found in their experiments that principals often preferred the less effective incentive contracts because they allowed the principals to appropriate a larger share of the surpluses produced from the principal-agent relationship even though the surpluses were, in total, smaller than those produced from fixed-wage contracts. Principals who stress the importance of a healthy work ethics rather than inherent owner interests, on the other hand, may see such efforts as complementing their compensation programs for the good of the firm, consistent with Murdock's (2002) argument that intrinsic motivation and incentive contracts could be complementary. The reason is that workers may perceive these actions as credible efforts by firm owners to improve organization efficiency rather than principal profits, thus encouraging agents to remain intrinsically motivated when explicit incentives are offered.

The model proposed in this paper advances the existing literature because it integrates the known psychology of motivation crowding out within a rational choice framework in which agents rationally respond to their perception of incentives by modifying their motivational orientations. The model is also consistent with much of the empirical evidence, and it produces a number of interested testable hypotheses. Clearly, however, more work is needed in modeling and explaining this important effect from an economic perspective. For example, the model developed in this paper treats the extrinsic and intrinsic components of agent utility as additive. This may not be consistent with some empirical findings that suggest the relationship between extrinsic and intrinsic motivation is interactive (Calder and Staw, 1975). Moreover, the model presented here does not fully conform to all empirical evidence. For example, it is well-known

that monetary compensation often reduces the incidence of blood donations. If one assumes that generalized norms govern the donation of blood, then according to the model developed in this paper compensation should not have a negative effect on the level of donations. Of course, it is possible that factors other than generalized norms are at work here and that the motivation to donate blood is not a generalized social norm but rather reflects an object that also supplies the rewards. But this would have to be explored in greater detail in order to understand and explain potential discrepancies between this model and any contrasting evidence. Therefore, economic models of agent motivation that more carefully describe the interrelationship between extrinsic rewards and intrinsic motivation consistent with the psychology of motivated behavior and the empirical evidence would greatly advance our understanding of motivation crowding out.

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Figures

Figure 1. Agent effort choices under two stylized scenarios. When incentives are zero ($r = 0$) and intrinsic motivation is present ($I = 1$), effort is $\hat{e}_1$. When incentives are offered ($r > 0$) and intrinsic motivation is absent ($I = 0$), effort is $\hat{e}_2$, which increases in the incentive rate, r . Observe that $e_1 \geq e_2$ when $\delta \geq r/p$.

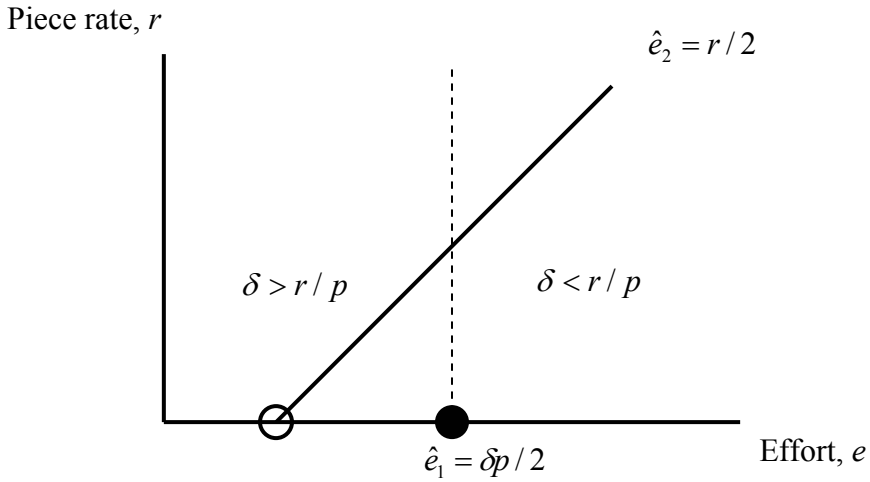


Figure 2. Optimal effort choices for an agent who is intrinsically motivated ($I = 1$) and is not intrinsically motivated ($I = 0$) under the two cases of no incentive compensation ($r = 0$) and positive incentive compensation ($r > 0$). When incentives are zero, the agent will choose cell III over cell I when $\delta \geq \delta_{FW}$. When incentives are positive, the agent will choose cell IV over cell II when $\delta \geq \delta_{IC}$.

	Incentive Absent $r = 0$	Incentive Positive $r > 0$
Intrinsic Motivation Absent ($I = 0$)	I $U = \bar{w}$ $\hat{e} = 0$	II $U = \frac{r^2 + 4\bar{w}}{4}$ $\hat{e} = r/2 \equiv \hat{e}_2$
Intrinsic Motivation Present ($I = 1$)	III $U = \frac{\delta^2 p^2 - 4\delta\bar{w} + 4\bar{w}}{4}$ $\hat{e} = \delta p/2 \equiv \hat{e}_1$	IV $U = \frac{\delta^2 (p-r)^2 + 2\delta(pr - r^2 - 2\bar{w}) + r^2 + 4\bar{w}}{4}$ $\hat{e} = [\delta(p-r) + r]/2$
Agent intrinsically motivated (i.e., $I = 1$) when	$\delta \geq \frac{4\bar{w}}{p^2} \equiv \delta_{FW}$	$\delta \geq \frac{4\bar{w} - 2r(p-r)}{(p-r)^2} \equiv \delta_{IC}$

Table 1. Stylized example of motivation crowding out, with sample effort choices and utilities of an agent for hypothetical values of the model's parameters. Compare the agent's utility and effort choice (columns 8 and 9, respectively) with and without intrinsic motivation (column 4) for different combinations of fixed and incentive compensation (e.g., compare row 1 to row 2, or row 11 to row 12). Bold indicates agent's "optimal" utility and effort for each comparison. Motivation crowding out occurs when the incentive rate, r , exceeds $\bar{r}$.

Row	Column								
	1	2	3	4	5	6	7	8	9
	$\bar{w}$	p	r	$I\delta$	δ_{FW}	δ_{IC}	$\bar{r}$	U	$\hat{e}$
1	300	70	0	0	0.24		60.23	300.00	0.00
2	300	70	0	.3	0.24		60.23	320.25	10.50
3	300	70	20	0		-0.32	60.23	400.00	10.00
4	300	70	20	.3		-0.32	60.23	516.25	17.50
5	300	70	50	0		-2.00	60.23	925.00	25.00
6	300	70	50	.3		-2.00	60.23	994.00	28.00
7	300	70	65	0		22.00	60.23	1356.25	32.50
8	300	70	65	.3		22.00	60.23	1315.56	33.25
9	600	70	0	0	0.49		47.30	600.00	0.00
10	600	70	0	.7	0.49		47.30	780.25	24.50
11	600	70	20	0		0.16	47.30	700.00	10.00
12	600	70	20	.7		0.16	47.30	936.25	27.50
13	600	70	50	0		1.00	47.30	1225.00	25.00
14	600	70	50	.7		1.00	47.30	1204.00	32.00
15	600	70	65	0		70.00	47.30	1656.25	32.50
16	600	70	65	.7		70.00	47.30	1353.06	34.25